Today

Linear Regression

Announcements

Lab 4 due Thursday

# Part V: Regression

**Regression:** Finding a model that fits data

$\hat{y} = m(x;\theta)$ — model, parameters, input, output prediction

parameters → collection of numbers, vectors, or matrices often written as one long vector

**Training / Fitting:**

$\underset{\theta}{\text{minimize}}\ J(\theta; X, Y)$ — total loss. Low $J$ means that the model fits the data well

$J(\theta)$ for short

$X, Y$: input data, output data — column vector or matrix

each row is a data point

n rows

Scalar: $X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$ Vector: $X = \begin{bmatrix} [\ \vec{x}_1\ ] \\ [\ \vec{x}_2\ ] \\ \\ [\ \vec{x}_n\ ] \end{bmatrix}$

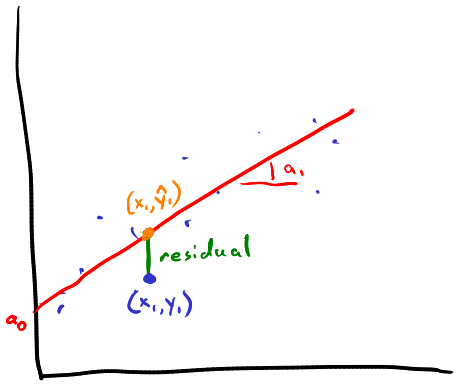
Typically $J(\theta; X, Y) = \frac{1}{n}\sum_{i=1}^{n} \ell(\hat{y}_i, y_i)$ — per sample loss

$= \frac{1}{n}\sum_{i=1}^{n} \ell(m(x_i;\theta), y_i)$

## Linear Regression (scalar)

$\hat{y} = m(x;\theta) = a_1 x + a_0$

$\theta = (a_0, a_1)$




$\ell(\hat{y}_i, y_i) = (\hat{y}_i - y_i)^2 = (a_1 x_i + a_0 - y_i)^2$

$\underset{\theta}{\text{minimize}}\ J(\theta; X, Y) \Rightarrow \underset{a_0, a_1}{\text{minimize}}\ \frac{1}{n}\sum_{i=1}^{n} (a_1 x_i + a_0 - y_i)^2$

F.O.N.C. $\nabla_\theta J(\theta^*; X, Y) = 0$ We can solve this analytically

$\vec{0} = \nabla_\theta J(\theta) = \begin{bmatrix} \frac{\partial J}{\partial a_0} \\ \frac{\partial J}{\partial a_1} \end{bmatrix}$ $J(\theta)$

$0 = \frac{\partial J}{\partial a_0} = \frac{1}{n}\sum_{i=1}^{n} 2(a_1 x_i + a_0 - y_i) = \frac{2}{n}\left(a_1 \Sigma x_i + n a_0 - \Sigma y_i\right)$

$0 = \frac{\partial J}{\partial a_1} = \frac{1}{n}\sum_{i=1}^{n} 2(a_1 x_i + a_0 - y_i) x_i = \frac{2}{n}\left(a_1 \Sigma x_i^2 + a_0 \Sigma x_i - \Sigma x_i y_i\right)$

$0 = a_1 \Sigma x_i + n a_0 - \Sigma y_i$

$0 = a_1 \Sigma x_i^2 + a_0 \Sigma x_i - \Sigma x_i y_i$

$\begin{bmatrix} n & \Sigma x_i \\ \Sigma x_i & \Sigma x_i^2 \end{bmatrix} \begin{bmatrix} a_0 \\ a_1 \end{bmatrix} = \begin{bmatrix} \Sigma y_i \\ \Sigma x_i y_i \end{bmatrix}$

using Cramer's rule

$$a_1 = \frac{n\Sigma x_i y_i - \Sigma x_i \Sigma y_i}{n\Sigma x_i^2 - (\Sigma x_i)^2}$$

$$a_0 = \frac{\Sigma y_i - a_1 \Sigma x_i}{n}$$

How do we quantify the accuracy of linear regression

$J(\theta)$: low = good, high = bad

problems:
- units of $J$ are units of $y^2$ (definition of "good" changes with each problem)
- ranges from $0 - \infty$

Common alternative: $r^2$ value "coefficient of determination"

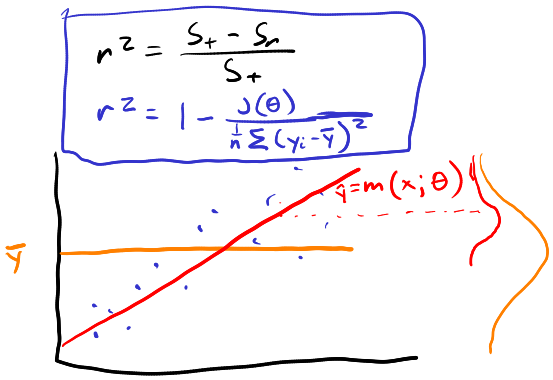
$$r^2 = \frac{S_t - S_r}{S_t}$$

$$r^2 = 1 - \frac{J(\theta)}{\frac{1}{n}\sum(y_i - \bar{y})^2}$$

$$S_r = \sum_{i=1}^{n}(\hat{y}_i - y_i)^2$$

$$S_t = \sum_{i=1}^{n}(y_i - \bar{y})^2$$

$$\bar{y} = \frac{1}{n}\sum_{i=1}^{n} y_i \quad \text{mean}$$

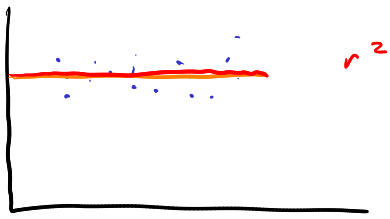



answers the question "how much better does the model explain the data than $\bar{y}$?"

ranges from 0-1 "percent of variance explained by the model"



$r^2 = 1 = 100\%$



$r^2 = 0$

highest $r^2$ value (best fit)

